

1.5 Compound Annual Growth Rate and Application Programming Interface



Calculate Compound Annual Growth Rate (CAGR) for Microsoft and Apple

CAGR summarises long-run growth as a single annualised rate. It is the **average annual rate at which an investment grows over time**, assuming the growth is compounded.

CAGR Formula:

$$\text{CAGR} = \left(\frac{P_T}{P_0} \right)^{\frac{1}{T}} - 1$$

where

- P_0 : Initial value
- P_T : Final value
- T: Number of years

Example

If a stock increases from 100 to 200 in 10 years, then

$$\text{CAGR} = \left(\frac{200}{100} \right)^{\frac{1}{10}} - 1 \approx 0.0718 = 7.18\%$$



Watch

Compound Annual Growth Rate

R Script (<https://learn.usc.edu.au/courses/35359/files/3224270?wrap=1>)_ 
(https://learn.usc.edu.au/courses/35359/files/3224270/download?download_frd=1)

Calculating Microsoft CAGR in R

```
msft_start <- microsoft$adjusted[1]
```

```
msft_end <- microsoft$adjusted[nrow(microsoft)]
```

```
msft_start_date <- microsoft$date[1]
```

```
msft_end_date <- microsoft$date[nrow(microsoft)]
```

```
msft_years <- as.numeric(difftime(msft_end_date, msft_start_date, units = "days")) / 365.25
```

```
msft_cagr <- (msft_end / msft_start)^(1 / msft_years) - 1
```

```
msft_cagr
```

Explanation:

- The first and last adjusted prices define the growth over the sample.
 - `difftime()` measures the time between the first and last date.
 - Dividing by 365.25 converts days into years.
 - The CAGR formula converts total growth into an annualised growth rate.
-

Now calculate Apple CAGR (using the same method)

```
aapl_start <- apple$adjusted[1]
```

```
aapl_end <- apple$adjusted[nrow(apple)]
```

```
aapl_start_date <- apple$date[1]
```

```
aapl_end_date <- apple$date[nrow(apple)]
```

```
aapl_years <- as.numeric(difftime(aapl_end_date, aapl_start_date, units = "days")) / 365.25
```

```
aapl_cagr <- (aapl_end / aapl_start)^(1 / aapl_years) - 1
```

```
aapl_cagr
```

Explanation:

- This is the same method applied to Apple.
-

Import Australian House Prices from FRED

Accessing External Data with an Application Programming Interface (API)

Before we use the `fredr` package, it is important to understand what an API is and why it is required.

What is an API?

An **API** is a structured way for one piece of software to communicate with another.

In practical terms:

- An API allows R to request data directly from an external database
- The database responds by sending the data back in a structured format
- This avoids manual downloads and helps ensure data is current and consistent

Many financial and economic data providers use APIs to manage access, monitor usage, and maintain data quality.

Why FRED Requires an API Key

FRED (Federal Reserve Economic Data) provides a large collection of economic and financial time series, including international indicators such as Australian house price indices.

To access FRED programmatically:

- You must register for a **free API key**
- The API key identifies you when you make data requests
- No payment or subscription is required

How to Obtain a FRED API Key

You only need to do this once.

1. Go to the [FRED website](https://fred.stlouisfed.org/docs/api/fred/).  (<https://fred.stlouisfed.org/docs/api/fred/>)
2. Create a free FRED account.
3. Navigate to your account settings and request an API key.
4. Copy the API key provided.

Setting Your API Key in R

Once you have your API key, you need to tell R what it is. This only needs to be done once per computer.

```
fredr_set_key("YOUR_API_KEY_HERE")
```

Explanation:

- `fredr_set_key()` stores your API key for future use
- Replace "YOUR_API_KEY_HERE" with your actual key
- You do not need to include the key again after this step

Download Australian House Price Data

Note: Ensure that you have installed and loaded the `fredr` package.

```
house_price <- fredr(  
  series_id = "QAUN628BIS",  
  observation_start = as.Date("2000-01-01")  
)
```

Explanation:

- `fredr()` downloads a time series from the FRED database
- `series_id = "QAUN628BIS"` specifies the Australian residential property price index
- `observation_start` sets the start date
- The returned table includes dates and a column called `value`

This series is an index, not a dollar price. However, growth rates and CAGR remain meaningful because they depend on proportional change.

Calculate House Price CAGR

We calculate CAGR using the same method as for stock prices.

```
house_start <- house_price$value[1]
```

```
house_end <- house_price$value[nrow(house_price)]
```

```
house_start_date <- house_price$date[1]
```

```
house_end_date <- house_price$date[nrow(house_price)]
```

```
house_years <- as.numeric(
```

```
  difftime(house_end_date, house_start_date, units = "days")
```

```
) / 365.25
```

```
house_cagr <- (house_end / house_start)^(1 / house_years) - 1
```

```
house_cagr
```

Explanation:

- The first and last index values measure total growth
- The time difference is converted from days to years
- The CAGR formula annualises growth over the sample period